

Djordje Miladinovic

Curriculum Vitæ

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Summary

- **My profile:** Machine learning researcher & engineer | PhD in Computer Science (AI-focused) | Swiss resident | Serbian citizen. 9+ years in machine learning research | 13+ years in software engineering | 4+ years in project leading.
- **My work:** Leading large-scale machine learning projects from concept to working system. This includes developing deep learning platforms for processing high-dimensional multi-modal data. My current focus is AI-driven drug discovery.

Experience



Machine Learning Researcher & Engineer, [GSK-AI Department](#)

Oct'21-Present

Applying machine learning to target identification in early drug discovery:

- Focus #1: Leading GSK efforts to build a deep learning platform for modeling genetic (CRISPR) perturbations.
- Focus #2: Developing machine learning platform for processing extremely high-dimensional multi-omics data.



Machine Learning Researcher, [ETH Zürich](#)

Feb'17-Jun'21

- **Sleep-learning web platform:** Led a team of researchers to develop a web-based large-scale machine learning platform for sleep pattern recognition from brain signals – *over 30K jobs processed to this date*.
- **SDN neural networks:** Co-developed a new type of neural network for realistic image synthesis. Used it to *develop a state-of-the-art variational autoencoder* for image modeling.
- **Zürich Exhalomics:** Led a team of students to develop a machine learning algorithm for interpretable identification of causal relations between metabolites and sleep stages (relating metabolism to sleep).



Machine Learning Researcher (Visitor), [Max Planck Institute for Intelligent Systems](#) *Sep'18-Dec'18*

- **Causal disentanglement framework:** Co-developed a neural architecture for learning disentangled representations.



Machine Learning Researcher & Engineer (Intern), [Logitech Europe S.A.](#)

Sep'16-Feb'17

- Developed a machine learning platform to target potential premium users to reduce churn rate.



Machine Learning Researcher (Visitor), [Disney Research Studios, Zurich](#)

Jan'16-Sep'16

- Developed a software platform for detecting mechanical failures in Disney's humanoid robots based on IMU sensor readings. The platform uses novel machine learning algorithm to assess robot degradation in a human-like way.

Education



PhD in Computer Science – Machine Learning, [ETH Zürich](#)

Sep'17-Jun'21

- **Thesis:** "On training Deep Generative Models with Latent Variables" – [Prof. Joachim M. Buhmann](#).
- **Research areas:** Generative image and text modeling | variational autoencoders | representation learning | deep learning | computer g | natural language processing | machine learning for biology.



MS in Computer Science – Machine Learning, [ETH Zürich](#)

Sep'13-Jun'16

- **Thesis:** "Perceptual Analysis Framework for Discovering Anomalies in Humanoid Motions" – [Prof. Otmar Hilliges](#).
- **Focus:** Machine learning | software engineering | operating systems | distributed computing.



BS in Computer Science – Software Engineering, [ETF Belgrade](#)

Sep'09-Sep'13

- **Thesis:** "Java-based Interactive Software Simulator of a CISC Processor" – [Prof. Zaharije Radivojević](#).
- **Focus:** Software engineering and design | algorithms | data structures | operating systems | hardware design.

Programming Languages

Advanced Python | PyTorch | Bash | $\LaTeX$
Intermediate C/C++ | Java | Matlab | SQL | HTML | CSS
Basic Tensorflow | Torch | Node.js | C# | Jekyll | Javascript | VHDL

Selected Publications

- [1] **Đorđe Miladinović**, Aleksandar Stanić, Stefan Bauer, Jürgen Schmidhuber & Joachim M. Buhmann
Spatial Dependency Networks: Neural Layers for Improved Generative Image Modeling
International Conference on Learning Representations, **ICLR 2021**
- [2] Raphael Suter, **Đorđe Miladinović**, Stefan Bauer & Bernhard Schölkopf
Robustly Disentangled Causal Mechanisms: Validating Deep Representations for Interventional Robustness
International Conference on Machine Learning, **ICML 2019**
- [3] **Đorđe Miladinović**, Muhammad Waleed Gondal, Bernhard Schölkopf, Joachim M. Buhmann & Stefan Bauer
Disentangled state space representations
DeepGen workshop, International Conference on Representation Learning, **ICLR 2019**
- [4] **Đorđe Miladinović** et al.
SPINDLE: End-to-end Learning from EEG/EMG to Extrapolate Animal Sleep Scoring Across Experimental Settings, Labs and Species, **PLoS Computational Biology 2019**

See the complete list at <http://djordjemila.github.io/#publications>

Academic Activities

Dec'19 Co-organized the "Disentanglement Challenge" at NeurIPS 2019 (bit.ly/36bTD4W).
2018-Present Associated research fellow of Max Planck Institute for Intelligent Systems.
2017-2021 Reviewed papers for NeurIPS, ICML and ICLR.
2017-2020 Taught "Advanced Machine Learning" and "Statistical Learning Theory" at ETH Zürich.
2017-2020 Supervised more than [10 MS students](#).

Languages

Serbian (Native), English (Fluent), German (Intermediate), Spanish (Beginner)

Hobbies

Waterpolo | Swimming | Skiing and snowboarding | Tennis | Cinematography | Reading